Forecasting and Analyzing Stock Prices Using Machine Learning Models

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PROJECT PLAN

PLAN	8 JUN	10 JUN	12 JUN	15 JUN	17 JUN	18 JUN

PHASE 1-3

95% — 8 JUN – 14 JUN

Data Specification & Pre-processing 100% — JUN 8 – JUN 10

Data Analysis & Integration 100% — JUN 10 – JUN 14

XGBoost Model 95% — JUN 10 – JUN 15

PHASE 4-6

Random Forest Model + Prophet Model — Jun 15 – JUN 18

Evaluation and Comparison

PROJECT STATUS UPDATE

- **Data Specification and Pre-processing**

Involves collecting, cleaning, transforming, and selecting relevant features from stock market data to prepare it for machine learning-based stock price prediction models.

- **Data Analysis and Integration**

involves examining and merging various data sources, performing statistical analysis, identifying patterns, and integrating the data into a cohesive format for machine learning-based prediction models.

- **Machine Learning Model 1**

The XGBoost Model is a powerful machine learning algorithm commonly used in stock price prediction projects, known for its strong performance and ability to handle complex patterns in data.

What's Left to Do!

Random Forest Model	Prophet Model	Evaluation and Comparison
- Fetch historical stock dataset - Feature Engineering by Select relevant features - Data Splitting - Random Forest Model Building - Prediction and Perform hyperparameters to optimize the Random Forest model's performance.	- Fetch historical stock dataset - Feature Engineering by Select relevant features - Data Splitting - Prophet Model Building - Prediction and Perform hyperparameters to optimize the Prophet model's performance.	- Select and identify Evaluation Metrics - Comparing models for performance evaluation. - Comparison and Interpretation of Results.

PROJECT IMPLEMENTATION

Demo..